2026-16-08

prove by induction for every $n \geq 1$ that the following holds true.

$$\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$$

First we prove a base case of $n = 1$, since 1 is the lowest value if $n \geq 1$

$$\sum_{i=1}^{1} i^3 = \left[\frac{1(1+1)}{2}\right]^2$$

$$1^3 = \left[\frac{2}{2}\right]^2$$

$$1 = 1$$

✓

Next we will prove that for any valid $n = k$, $n = k+1$ holds true. First, we assume that for any valid $n = k$, our equation holds true.

$$\sum_{i=1}^{k} i^3 = \left[\frac{k(k+1)}{2}\right]^2$$

Now we prove that $n = k+1$ holds true using our assumed $n = k$.

$$\sum_{i=1}^{k+1} i^3 = \left[\frac{(k+1)(k+2)}{2}\right]^2$$

$$\sum_{i=1}^{k} i^3 + (k+1)^3 = \left[\frac{(k+1)(k+2)}{2}\right]^2$$